Keynty CMOS CW Keyer

Schematic and design-intent review

Project	`github.com/vash909/keynty`, schematic rev. 0.0 (2026-04-03)
Commit	`2fe49df` (“added LCSC number for integrated circuits, leds and capacitors”)
Scope	Schematic, component choices and design intent. PCB layout excluded.
Date	1 November 2026
Reviewer	Claude (Anthropic)

Verdict. As drawn, Keynty will produce plain dits and a single dah started from rest, so it can pass a quick bench test, but it will **not work as an iambic keyer**. Two logic faults break dah generation and the dot/dah memory (F1, F2). The way the 4027 clock inputs are driven (F3) and a 3 V supply below the specified range of the chosen 4000B parts (F4) make the flip-flops marginal on top of that. All four are fixable without changing the overall architecture.

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1 Summary of findings

ID	Severity	Finding	Effect on operation
F1	CRITICAL	Dah divider U3B and selector U3A are clocked by the same edge; U3B reads the selector's *old* output	A dah that follows a dit while the keyer is running comes out as a dit; squeeze produces only dits
F2	CRITICAL	Memory latches are not OR-ed into the oscillator enable	Memorised elements are never sent; Mode B does not exist; stale memory corrupts the next element
F3	HIGH	Slow, loaded clock edges on both 4027 clock pins (C8 clamped via 100 Ω; Q1 gate on OUT)	Double-clocking risk; the from-rest DOTS edge is hundreds of µs long
F4	HIGH	2×AA supply below the 3 V minimum of the 4000B parts; 33 Ω in series with all loads	Logic out of spec for most of battery life; VCC dips 0.5–1 V on every key-down with sidetone
F5	MEDIUM	No power-on reset on the 4027	Spurious key-down (≈2 dit lengths) at switch-on in some power-ups
F6	MEDIUM	DOT/DASH nets only pulled down through the J2/J3 jumpers	Missing or mis-set jumper leaves CMOS inputs floating
F7	LOW	Paddle common is +VCC on the jack sleeve	Works while fully floating; traps if the paddle is ever shared or grounded
F8	LOW	IAMBIC A/B labels on SW2 swapped	Wrong mode label once F2 is fixed
F9	MEDIUM	PC817 driven at ≈2–3.8 mA; unidirectional output	Marginal key-line sink current on some rigs; no negative keying
F10	LOW	Power LED draws ≈5 mA continuously	Contradicts the "reduced consumption" goal; dominates battery drain
F11	LOW	Oscillator formula on the schematic does not apply; speed range and taper	Speed ≈25 % slower than calculated; drifts with battery
F12	LOW	Sidetone depends on BZ1 being an active (self-oscillating) buzzer	A passive transducer will only click

Recommended order of work: **F2** (cheap, independent of timing), then **F1 and F3 together** (they interact: fixing R7 alone makes F1 worse), then **F4**, then the rest.

2 Scope and method

- **Netlist extraction.** The netlist was extracted directly from `keynty_keyer.kicad_sch` (KiCad 10 format) with a purpose-written parser (symbol pin transforms, wires, junctions, labels, power symbols). It was cross-checked against the pad nets stored in `keynty_keyer.kicad_pcb`: connectivity is identical. The only discrepancy is that the PCB still carries the placeholder value "C" for C6/C7 (220 nF in the schematic), i.e. the board was not re-synced after the last value change. The full logic netlist is in Appendix A.
- **Behavioural model.** The logic was modelled exactly as wired (Section 7, source in Appendix B), including a normalised analogue model of the gated oscillator and its clamp.
- **Reference design.** The original Minty "Kool" Keyer by K4ICY [1] was used to understand design intent and to identify what changed.
- **Parts.** The LCSC numbers in the BOM were looked up where possible [2, 3].

The PCB layout itself was not reviewed, as requested.

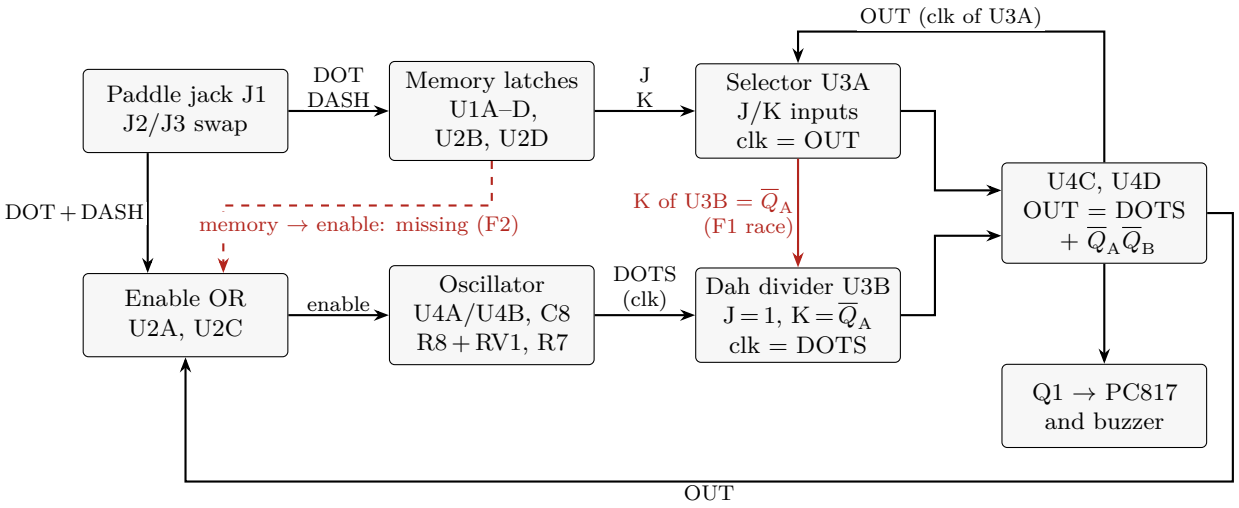


Figure 1: Signal flow of the keyer logic as wired. Red: the two critical faults.

3 How the circuit works (as wired)

Inputs. Paddle tip and ring (J1) are pulled to GND by R4/R3 (10 k) with C7/C6 (220 n); the paddle common (sleeve) is tied to +VCC, so the inputs are active-high. J2/J3 route tip/ring to the DOT and DASH nets (paddle swap).

Oscillator. U4B is a gated NAND (U4 pin 5 from the timing node via R7, U4 pin 6 = enable), U4A an inverter. C8 (220 n) couples DOTS (U4 pin 3) back to the timing node; R8 (150 k) + RV1 (1 M) return it to $\overline{\text{DOTS}}$ (U4 pin 4). When enabled the oscillator starts immediately, and after a stop it cannot restart until the node has recovered, which gives a self-completing inter-element space.

Selector. U3A (pins 1–7) is clocked by OUT. J = dit request (DOT or dit memory), K = dah request (DASH or dah memory). $\overline{Q}_A$ high means "dah selected".

Dah generator. U3B (pins 9–15) is clocked by DOTS with J = 1 and K = $\overline{Q}_A$. When a dah is selected it divides by two, so $\overline{Q}_B$ stays high for the second half-period and OUT = DOTS + $\overline{Q}_A\overline{Q}_B$ is held for three units.

Memory. U1D/U1C/U2B latch a dit request while a dah is selected; U1A/U1B/U2D latch a dah request while a dit is selected. MODE (SW2) enables the latches permanently (VCC) or only while a paddle is held (DOT + DASH).

Enable. U4 pin 6 = OUT + DOT + DASH (U2A, U2C).

Output. OUT drives Q1's gate directly. Q1 sinks the PC817 LED (R9 = 470 Ω) and, via SW3, the buzzer.

Supply. 2×AA → SW1 → R1 (33 Ω) → VCC; 1 µF decoupling per IC; power LED via R2 (220 Ω).

What is right. The gated-oscillator topology (immediate start, self-completing space) is sound and inherited correctly. The three-unit dah logic is correct, and OUT is glitch-free at the mid-dah edge because DOTS rises before $\overline{Q}_B$ falls. Every gate in U1, U2 and U4 is used, so there are no floating spare inputs, and the 4027 set/reset pins are tied. Each IC has local decoupling. The paddle RC debounces contact bounce on make (the capacitor charges instantly and decays in ≈2.2 ms), and the opto-isolated output keeps the keyer floating relative to the rig.

4 Critical findings

F1 The dah divider reads the selector one element late

CRITICAL

Where. U3A (selector) is clocked by OUT on U3 pin 3; U3B (dah divider) is clocked by DOTS on U3 pin 13; U3B's K input (U3 pin 11) is U3A's $\overline{Q}_{\mathrm{A}}$ (U3 pin 2).

Mechanism. At the start of every element, both clocks are derived from the same falling edge of $\overline{\mathrm{DOTS}}$ (U4 pin 4), one gate later each: U4D produces OUT and U4A produces DOTS. U3B samples K at its clock edge (the 4027's hold time is essentially zero), but U3A's output only changes after its own clock-to-Q delay, which is several hundred nanoseconds for a 4000B part at 3 V. U3B therefore decides "dit or dah" with the *previous* element's selection (Figure 2a).

Consequence. Whenever the oscillator is already running, a dah that follows a dit is generated as a dit, because U3B is set rather than toggled. Squeezing both paddles produces a continuous string of dits after the first element. An "A" sent at normal speed comes out as two dits (or "U" if the dah paddle is held).

Why it can look fine on the bench. At rest, the timing node (C8/R7) sits at VCC. When the first DOTS edge starts, the node is pushed above VCC and is clamped by the protection diode of U4 pin 5 through only 100 Ω. From then on U4A has to charge 220 nF with its limited output current, so DOTS stalls just above 0.5 V and then ramps slowly through the clock threshold over hundreds of microseconds (Figure 2b). By the time U3B is clocked, U3A has long settled. This accident makes the first element after a pause correct, and only that element. It also means that "fixing" R7 on its own (F3) makes things worse: the model then turns even a lone dah after a dit into a dit.

Corroboration. K4ICY's write-up of the original design [1] describes the same class of symptom (random elements, or a keyer that only produces dits) and attributes it to the 4027 clock inputs. His mitigation, small capacitors on the clock pins, worked on his build but not on others', which is what a race resolved by parasitic timing looks like.

Fix. Guarantee that U3B samples after U3A has settled, by design rather than by parasitics:

(a) Delay U3B's clock: DOTS → 10 k → 1 nF to GND → Schmitt buffer → U3 pin 13. $\tau = 10\,\mu s$ is far longer than U3A's clock-to-Q even at low VDD and far shorter than a dit (≥ 25 ms). The Schmitt buffer (e.g. a single-gate 74LVC1G17, or a 40106 section) restores a fast edge for the 4027. No spare gate exists in U1, U2 or U4, so this needs one extra part.

(b) Or remove the dependency: drive U3B's K from the selector's *next* state instead of its present output. With $Q_A = 1$ meaning "dit", the 4027 next state is $Q_A^+ = J\overline{Q}_A + \overline{K}Q_A$, so $K_B = \overline{J\overline{Q}_A + \overline{K}Q_A}$. This costs about four gates but has no timing dependency.

Do this together with F3, and verify on a scope as described in Section 8.

F2 Dot/dah memory never restarts the oscillator

CRITICAL

Where. The oscillator enable on U4 pin 6 is U2A = OUT + (DOT + DASH) from U2C. The memory nets, J of U3A (U2 pin 4) and K of U3A (U2 pin 11), are not part of it.

Design intent. In the original Minty Keyer, the paddle inputs, the output and the memory outputs all feed the oscillator enable through a diode OR [1]. The README states the diode OR was replaced by CMOS OR gates; in the process the memory inputs were dropped.

Consequence. If the opposite paddle is tapped and released before the current element and its following space finish, the latch is set but the oscillator stops at the end of the space, and the memorised element is never sent. Iambic Mode B therefore does not exist; the keyer behaves like a no-memory keyer at best. Worse, with MODE tied to VCC the latch stays set while idle, so it corrupts the *next* press: send a dah, tap the dit paddle during it, pause, then tap the dah paddle, and the keyer sends "T E" instead of "N T" (the stored dit makes U3A toggle to dit).

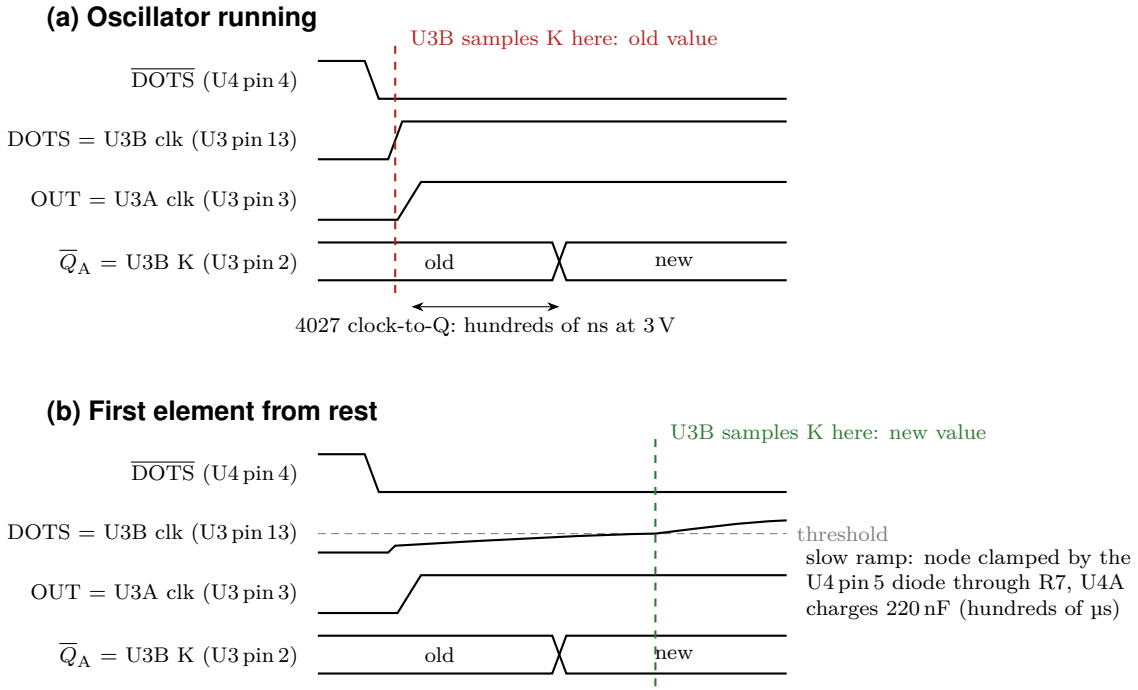


Figure 2: The F1 race at the start of an element (qualitative, not to scale; panel (b) has a much longer time axis). (a) While the oscillator runs, the divider's clock edge arrives before the selector's output has changed. (b) From rest, the clamped timing node slows the DOTS edge so much that the selector wins. This is accidental, and the edge itself violates the 4027's clock rise-time limit (F3).

Fix. Make the enable OUT + J-net + K-net. The J and K nets already contain the paddle signals, so they can replace DOT + DASH at U2 pin 2: feed U2 pin 2 from a two-diode OR of U2 pin 4 and U2 pin 11 (e.g. BAT54C with a 100 k pull-down), or use a 3-input OR such as a 4075. Keep U2C (DOT + DASH) for the MODE switch. The model confirms that with this change the memory and both iambic modes behave correctly (Section 7).

5 High-severity findings

F3 Both 4027 clock inputs are driven by slow, loaded edges HIGH

DOTS (U3 pin 13). DOTS drives C8 (220 nF) into a timing node that is clamped by the input protection diodes of U4 pin 5 through only R7 = 100 Ω. Every edge therefore has a slow tail while U4A charges C8 through the clamp, and from rest the whole edge is slow (F1, Figure 2b). The 4027 specifies a maximum clock rise/fall time in the microsecond range; an edge that takes hundreds of microseconds to cross the threshold invites double toggling of U3B, which shows up as one-unit or five-unit "dahs". The clamp current also flows through the input protection network of U4 on every half-cycle. It is limited by U4A's weak output so it stays within the absolute rating, but it is poor practice in a 4000B design.

OUT (U3 pin 3). OUT also drives Q1's gate directly. During turn-on, the gate voltage sits on the MOSFET's Miller plateau, roughly 1–1.5 V, which on a 3 V supply is right at the 4027's clock threshold. At the same moment the buzzer current step is pulling VCC down through R1 (F4). That is a classic recipe for double-clocking the selector, which would make iambic sequences skip or repeat elements. The original drove its keying MOSFETs through a 10 k resistor, which isolates the logic node from the gate charge.

Fix.

- Raise R7 to ≥ 1 MΩ (the usual rule for this astable is a series resistor several times the timing resistor), so the node is no longer hard-clamped. Only do this together with F1. The period

then becomes roughly $2.2RC$ instead of the clamped value, so rescale C8 or R8 to keep the speed range (F11).

- Alternatively use a 4093 (Schmitt NAND, pin-compatible with the 4011) for U4 and give the 4027 clocks a clean, buffered edge.
- Drive Q1's gate through 1–10 k so that only U4D's own output, not the MOSFET's gate charge, sits on the U3A clock net.

F4 Supply voltage below the logic's specified range, and a 33 Ω series resistor HIGH

Logic supply range. Two alkaline AA cells give about 3.2 V fresh, about 2.4 V at mid-life and about 2.0 V at end of life. The BOM's LCSC numbers for the 4011 and 4071 resolve to Nexperia HEF4011BT and HEF4071BT [2, 3], standard 4000B parts specified from 3 V. The logic therefore spends most of the battery life below its minimum specified supply. At 2–2.5 V, 4000B gates usually still switch, but output drive collapses and propagation delays grow to the microsecond range, which makes F1 and F3 worse. The README's claim that the gates were changed to low-voltage (3.6 V max) parts does not match the BOM. Note also that 74HC/74LVC gates cannot simply be substituted: their pinout differs from the 4000 series for gates B and C (4011/4071/4081 gate B is in 5,6 / out 4; 74xx00/32/08 gate 2 is in 4,5 / out 6).

Series resistor R1. R1 (33 Ω) carries every load:

- Power LED: ≈5 mA continuously → 0.17 V drop.
- Buzzer: a 12 mm magnetic buzzer typically draws 20–30 mA → a further 0.7–1 V drop on every key-down with sidetone on. The 5 µF of total decoupling with 33 Ω has $\tau \approx 165$ µs, so VCC stays low for the whole mark.
- PC817 LED: ≈3.8 mA while keyed.

The dip lands exactly when the flip-flops are clocked (F3), shifts the oscillator timing between sidetone on and off, and reduces the opto drive (F9).

Fix. Run the logic at 4.5 V (3×AA) or 9 V as the original did; 4000B parts are comfortable there and the opto and LED resistors simply scale. Replace R1 with 0 Ω (or ≤ 2 Ω if a fuse-like element is wanted), add 47–100 µF of bulk capacitance, and choose a low-current active piezo buzzer (≈5 mA) or feed the buzzer from the battery side of R1.

6 Medium and low-severity findings

F5 No power-on reset MEDIUM

The 4027 set/reset pins (U3 pin 4, 7, 9, 12) are tied to GND and the memory latches have no defined state. If both $\overline{Q}_A$ and $\overline{Q}_B$ come up high, OUT is high immediately; the keyer stays key-down until the first DOTS edge toggles U3B, about two dit lengths. Connected to a rig, it transmits when switched on. Fix: an RC power-on pulse on U3 pin 9 (S of U3B; e.g. 100 nF from VCC, 100 k to GND) forces $\overline{Q}_B$ low so OUT cannot start high. Once F2 is applied, a memory latch that powers up set will send one element at switch-on; gate MODE with the same pulse if that matters.

F6 Paddle inputs can float MEDIUM

R3/R4 pull down the *jack* side. The logic-side DOT and DASH nets (U2 pin 5, 8, 9, 12) are only pulled down through the J2/J3 jumpers. A missing jumper, a loose one, or both headers set to the same position leaves a CMOS input floating, which causes random keying and extra supply current. Fix: add 100 k pull-downs directly on the DOT and DASH nets, or replace J2/J3 with a DPDT slide switch.

F7 Paddle common is +VCC on the sleeve LOW

This works because the keyer is battery-powered and opto-isolated, so its ground floats relative to everything else. It is unconventional, though: anything that ties the paddle common to the keyer's ground, such as a shared paddle on a Y-cable to the rig's own key input, a grounded metal case, or a later move to an external supply shared with the rig, either shorts the supply through R1 or makes the inputs fight. At minimum, document it on the silkscreen; the conventional alternative is a grounded common with pull-ups and an inversion stage, as in the original.

F8 IAMBIC A/B labels on SW2 are swapped LOW

In the schematic, the position that ties MODE to VCC (memory independent of the paddles, i.e. Mode B behaviour) is labelled "IAMBIC A", and the DOT + DASH position (memory cleared when both paddles are released, i.e. Mode A) is labelled "IAMBIC B". This only matters once F2 is fixed.

F9 Opto-coupler drive is thin and unidirectional MEDIUM

With R9 = 470 Ω, the PC817 LED current is ≈3.8 mA on fresh cells and ≈2 mA at 2.2 V (end of life, or during a buzzer dip). With the PC817's minimum CTR of 50 % this guarantees only about 1 mA of collector current, with saturation not guaranteed at that ratio. Most modern rigs pull their key line up with ≤ 1 mA and will work; some will not. The phototransistor also conducts in one direction only (tip must be positive), whereas the original used back-to-back MOSFETs for bidirectional keying. Fix: R9 = 150–220 Ω (current flows only while keyed); for bidirectional or higher-voltage keying use a PhotoMOS relay, as the original's keying-interface option did [1].

F10 The power LED dominates consumption LOW

R2 = 220 Ω with a red LED draws ≈5 mA continuously, orders of magnitude more than the idle CMOS logic. That alone limits an alkaline AA pair to a few hundred hours of power-on time and contradicts the README's "reduced overall power consumption". Fix: high-efficiency LED with 2.2–4.7 k (≈0.2–0.5 mA), or a momentary battery-check button.

F11 Oscillator formula, speed range and pot taper LOW

The schematic note gives $T = 2 \ln 2\, RC$. With the timing node clamped by a diode, each half-period is instead

$$t_{\text{half}} \approx RC \, \ln \frac{V_{DD} + V_D}{V_{DD}/2} \approx 0.87\, RC \quad (V_{DD} = 3\,\text{V},\ V_D \approx 0.6\,\text{V}),$$

about 25 % longer than $\ln 2\, RC$, and it grows as the battery discharges (≈0.92 RC at 2.4 V, ≈0.96 RC at 2.0 V). With C8 = 220 nF and R = 150 k … 1.15 M this gives roughly 41 down to 5 WPM. Because speed is proportional to $1/R$, a linear 1 M pot crowds the useful 15–30 WPM range into a small part of the rotation; a reverse-log pot or a smaller pot with a series resistor gives a more usable control.

F12 Sidetone relies on an active buzzer LOW

Q1 switches DC through BZ1, so the sidetone only works if BZ1 is an active (self-oscillating) buzzer rated for ≤ 3 V. A passive 12 × 9.5 mm transducer on the same footprint will only click. The LCSC number in the BOM (C49246930) could not be verified for this review. If a magnetic type is used, add a flyback diode across it; its inductive kick otherwise also appears at the PC817 LED cathode.

7 Behavioural simulation

Model. The logic is modelled exactly as wired (Appendix B). The oscillator node is modelled in normalised analogue form ($V_{DD} = 3$ V, threshold $V_{DD}/2$, clamp diodes at −0.6 V and V_{DD} + 0.6 V), so the self-completing space and the slow first edge emerge from the circuit rather than being imposed. Inputs are paddle press/release times in dit units; the output is the sequence of marks on OUT.

Assumptions. (1) While the oscillator runs, U3B samples U3A's old output (F1 mechanism). (2) When DOTS starts from a node voltage high enough that it hits the clamp before the clock threshold, U3A settles first. (3) Supply dips and double clocking (F3, F4) are *not* modelled, so real hardware can only be less regular than shown. The "race fix" variant forces correct ordering; the "memory fix" variant adds the J/K nets to the oscillator enable. MODE is tied to VCC (Mode B behaviour).

Paddle input	Correct	As built	F1 fix only	F2 fix only	F1 + F2
Hold dit paddle	▪▪▪	▪▪▪	▪▪▪	▪▪▪	▪▪▪
Tap dah from rest (last element a dit)	▬	▬	▬	▬	▬
"A", dah pressed during the dit	▪▬	▪▪	▪▬	▪▪	▪▬
"A", dah pressed in the following gap	▪▬	▪▪	▪▬	▪▪	▪▬
"A", dah pressed >1.5 units after the dit	▪▬	▪▬	▪▬	▪▬	▪▬
Squeeze both, dit first, release	▪▬▪▬▪	▪▪▪▪▪▪	▪▬▬▪▬▬	▪▪▪▪▪▪▪	▪▬▬▪▬▪
Squeeze both, dah first, release	▬▪▬▬▪▬	▬▬▬▪▪▪	▬▬▪▬▬▪	▬▬▬▪▪▪	▬▬▪▬▬▪
"N", dit tapped and released during the dah	▬▪	▬	▬	▬▪	▬▪
"N" as above, pause, then "T"	▬▪▬	▬▪	▬▪	▬▪▪	▬▪▬

Table 1: Element sequences produced on OUT for typical inputs (Mode B). Red: incorrect output. Neither fix alone is sufficient; with both, every case is correct.

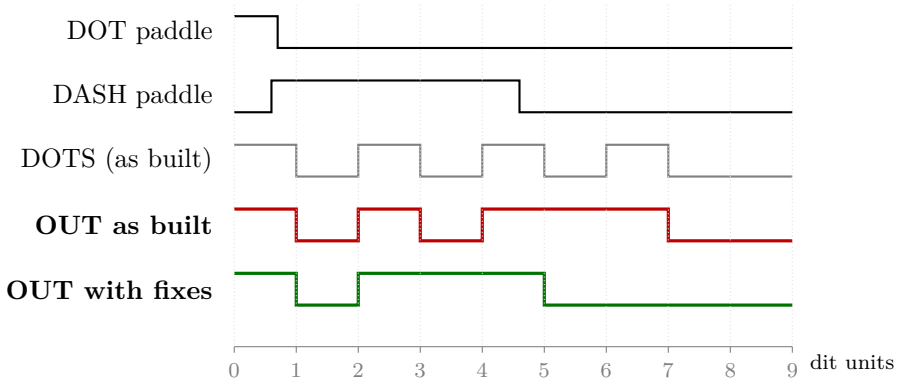


Figure 3: Sending "A": the dah paddle is pressed while the dit is still being sent. As built, the second element is a dit ("U" is sent instead of "A"); with F1 and F2 fixed the output is correct. Grid in dit units.

8 Recommended changes and verification

Changes, in order

1. **F2**: add the J/K memory nets to the oscillator enable (two diodes and a resistor, or a 3-input OR).
2. **F1 + F3**: guarantee the U3A-before-U3B ordering (delayed, Schmitt-buffered U3B clock, or next-state K); raise R7 and rescale the timing; drive Q1's gate through a resistor.
3. **F4**: 4.5 V or 9 V supply, R1 to 0 Ω, bulk capacitor, low-current buzzer.
4. **F5, F6**: power-on pulse on U3 pin 9; pull-downs on the logic-side DOT/DASH nets.
5. **F7–F12**: labels, opto drive, LED current, pot taper, buzzer type.

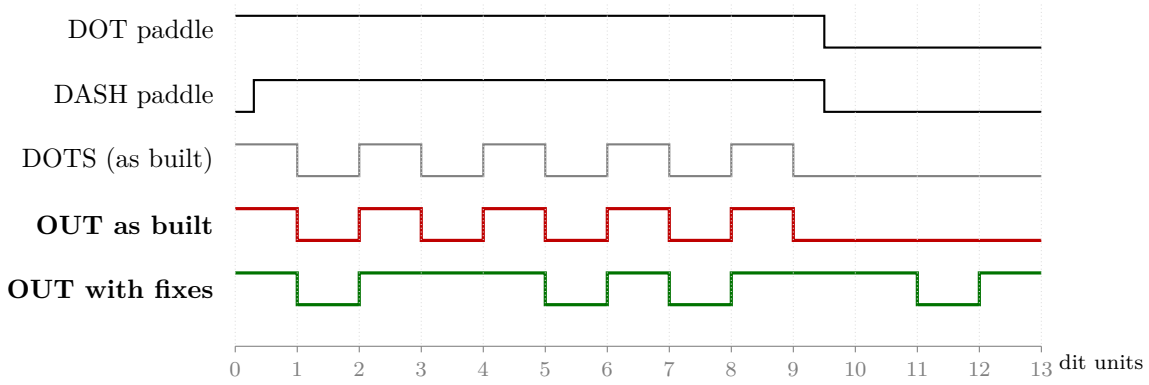


Figure 4: Squeeze, dit paddle first, both released at 9.5 units. As built, only dits are produced. With F1 and F2 fixed, the keyer alternates and, in Mode B, adds the trailing opposite element after release.

Bench checks that confirm or refute each finding

- **F1:** scope CH1 on U3 pin 13, CH2 on U3 pin 2, trigger on CH1 rising while squeezing. If U3 pin 2 changes *after* the CH1 edge on element changes, the race is confirmed; on the audio, squeeze gives only dits.
- **F2:** hold the dah paddle, tap and release the dit paddle during the dah. No dit follows: confirmed.
- **F3:** zoom on U3 pin 13 for the first element after a pause: a slow ramp lasting hundreds of µs. Probe U3 pin 3 during key-down for a plateau near VDD/2.
- **F4:** AC-coupled probe on VCC with sidetone on: expect a 0.5–1 V dip for every mark.
- **F5:** power-cycle with the output connected to an LED or a dummy load: watch for a spurious key-down.

A Logic netlist (extracted from the schematic)

Net	Connections
VCC	R1.2, C1–C5, J1 sleeve, R2.1, R9.1, SW2.A, SW3.A, U1.14, U2.14, U3.16, U3.10 (J of U3B), U4.14
GND	C1–C7, D1.K, J6, Q1.S, R3.1, R4.1, U1.7, U2.7, U3.8, U3.4/7/9/12 (all R/S), U4.7
Battery +	J5 → SW1 → R1 (33 Ω) → VCC
Paddle tip	J1.T, J2.2, R4 (10 k to GND), C7 (220 n to GND)
Paddle ring	J1.R, J3.2, R3 (10 k to GND), C6 (220 n to GND)
DOT	J2.3, J3.3, U2.5, U2.8
DASH	J2.1, J3.1, U2.9, U2.12
DASH+DOT	U2.10 (U2C out), U2.2, SW2.C
MODE	SW2.B (common), U1.6, U1.8
J-net (U3A J)	U2.4 (U2B out), U3.6, U1.13
K-net (U3A K)	U2.11 (U2D out), U3.5, U1.1
U3A Q	U3.1, U1.2
$\overline{Q}_A$ (DASH/$\overline{\text{DOT}}$)	U3.2, U3.11 (K of U3B), U1.12, U4.8
Memory loop	U1.3 → U1.5; U1.4 → U2.13; U1.11 → U1.9; U1.10 → U2.6
Enable	U2.3 (U2A out) → U4.6
$\overline{\text{DOTS}}$	U4.4 (U4B out), U4.1, U4.2, U4.13, R8.2
DOTS	U4.3 (U4A out), C8.2, U3.13 (clk of U3B)
Timing node	C8.1, RV1.1, RV1.2, R7.1; R7.2 → U4.5; R8.1 – RV1.3
$\overline{Q}_B$	U3.14, U4.9 (U3.15 not connected)

Net	Connections
$\overline{\text{DASHES}}$	U4.10, U4.12
OUT	U4.11, U3.3 (clk of U3A), U2.1, Q1.G
Keying	Q1.D, BZ1−, U5.2 (LED cathode); BZ1+ ← SW3 ← VCC; U5.1 ← R9 (470 Ω) ← VCC
Key output J4	J4.T → U5.4 (collector); J4.R, J4.S → U5.3 (emitter)
Power LED	VCC → R2 (220 Ω) → D1 → GND

Parts: U1 4081 (LCSC C52342), U2 4071 (C28869, HEF4071BT), U3 4027 (C157536), U4 4011 (C6725, HEF4011BT), U5 PC817, Q1 ZXM61N02F (BSS138 symbol), C8 220 nF, R8 150 k, RV1 1 M, R7 100 Ω, R1 33 Ω.

B Behavioural model source

```
"""Behavioural model of the Keynty keyer logic, exactly as wired in keynty_keyer.kicad_sch (commit 2fe49df).

Oscillator : U4B (gated NAND) + U4A (inverter), C8 = 220 nF, R8 + RV1 timing, R7 = 100 R into U4 pin 5 clamp diodes.
Selector   : U3A (J = DOT | dot-memory, K = DASH | dash-memory), clocked by OUT.
Dah divider: U3B (J = 1, K = U3A /Q), clocked by DOTS.
Output     : OUT = DOTS | (/Q1 & /Q2)     (U4C, U4D)
Memory     : J = DOT  | (MODE & /Q1 & J)  (U1D, U1C, U2B)
             K = DASH | (MODE &  Q1 & K)  (U1A, U1B, U2D)
Enable     : U4 pin 6 = OUT | DOT | DASH  (U2A, U2C)  -- memory NOT included (as built)

Normalised analogue model of node X (C8/R7 node): VDD = 3 V, gate and clock threshold = VDD/2,
RC = 1, clamp diodes at -0.6 V and VDD + 0.6 V.  One dit = RC*ln((VDD+0.6)/(VDD/2)).

race = 'physical': U3B samples U3A's OLD /Q (both flip-flops clocked by the same edge), except when
                   DOTS starts from rest with node X above VDD+0.6-VDD/2, where the clamp makes the DOTS
                   edge slow and U3A settles first.
race = 'new'     : ordering guaranteed (proposed fix).
mem_in_enable    : memory nets OR-ed into the oscillator enable (proposed fix).
"""
import math
VDD, VT, VD = 3.0, 1.5, 0.6
UNIT = math.log((VDD + VD) / VT)

def sim(press, T=20, mode='B', race='physical', mem_in_enable=False, dt=0.0005, trace=False):
    t = 0.0; X = VDD; Q1 = 1; Q2 = 1          # idle after a dit
    J1n = K1n = 0; OUT = 0; DOTS = 0; nDOTS = 1
    hist = []
    def paddles(t):
        d = any(a <= t < b for a, b in press.get('dot', []))
        h = any(a <= t < b for a, b in press.get('dash', []))
        return int(d), int(h)
    while t < T * UNIT:
        DOT, DASH = paddles(t)
        MODE = 1 if mode == 'B' else (DOT | DASH)
        nQ1 = 1 - Q1
        for _ in range(4):
            J1n = DOT | (MODE & nQ1 & J1n)
            K1n = DASH | (MODE & Q1 & K1n)
        EN = OUT | DOT | DASH | ((J1n | K1n) if mem_in_enable else 0)
        new_nDOTS = 0 if (X > VT and EN) else 1
        if new_nDOTS != nDOTS:
            if new_nDOTS == 0:                      # DOTS rising
                slow = X > (VDD + VD - VT)
                fast_race = (race == 'old') or (race == 'physical' and not slow)
                oldnQ1 = nQ1
                if OUT == 0:                        # OUT rises -> U3A clocked
                    if J1n and K1n: Q1 = 1 - Q1
                    elif J1n: Q1 = 1
                    elif K1n: Q1 = 0
                k2 = oldnQ1 if fast_race else 1 - Q1
                Q2 = (1 - Q2) if k2 else 1
                X = min(X + VDD, VDD + VD)
            else:
```

```
                X = max(X - VDD, -VD)
            nDOTS = new_nDOTS; DOTS = 1 - nDOTS
        X += ((VDD if nDOTS else 0.0) - X) * dt
        OUT = DOTS | ((1 - Q1) & (1 - Q2))
        hist.append((t / UNIT, DOT, DASH, DOTS, OUT))
        t += dt
    marks = []; run = 0; prev = 0
    for h in hist + [(0, 0, 0, 0, 0)]:
        o = h[4]
        if o: run += 1
        elif prev: marks.append(run * dt / UNIT); run = 0
        prev = o
    s = ''.join('.' if m < 2 else '-' for m in marks)
    return (s, marks, hist) if trace else (s, marks)

def units(p):
    return {k: [(a * UNIT, b * UNIT) for a, b in v] for k, v in p.items()}
```

References

[1] K4ICY, *Minty "Kool" Keyer — 3-IC electronic CW keyer*, https://www.k4icy.com/weekend_radio_minty_kool_keyer.html.

[2] LCSC, *HEF4011BT,653 (C6725)*, https://www.lcsc.com/product-detail/Gates_Nexperia_C6725.html.

[3] LCSC, *HEF4071BT,653 (C28869)*, https://www.lcsc.com/product-detail/Gates_Nexperia-HEF4071BT-653_C28869.html.

[4] Texas Instruments, *CD4027B CMOS dual J-K master-slave flip-flop*, datasheet SCHS032, https://www.ti.com/product/CD4027B.

[5] vash909, *keynty*, https://github.com/vash909/keynty, commit 2fe49df.